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# Exploring the size ratio impact on an intruder segregating in bedload transport

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Vertical size segregation in bedload transport impacts the sediment rate and the river bed morphology. To better account for this process in sediment transport models, it is essential to understand the mechanisms acting at the grain scale. This paper focuses on the behavior of a single large particle called the intruder, segregating upwards in a granular bed made of smaller mono-disperse beads, during bedload transport. Experiments were carried out in a narrow flume, where the distance between the sidewalls was of the order of the intruder diameter. The latter was initially buried a few diameters below the bed surface. While supplying the flume with constant rates of small beads and water, we used a high speed camera and image analysis techniques to monitor both the trajectory of the intruder and the velocity of the granular bed. We performed approximately 300 experiments with six size ratios (large to small) ranging from 1.25 to 3.0 to get statistically representative intruder trajectories. This extensive and unusual dataset allows for a more thorough validation of the observations made in the work of Rousseau *et al.* [*Phys. Rev. Fluids* **7**, 064305 (2022)]. Specifically, our results highlight the transition in the intruder behavior occurring around a size ratio of 2. Below this value, the intruder spatial trajectory exhibits a greater variability and gets steeper as the size ratio increases. Above the value of 2 and whatever the size ratio, the spatial trajectory is almost linear with a constant slope and the time taken by the intruder to reach the bed surface increases with the size ratio. In addition, our extensive dataset reveals new insights into the intruder kinematics. Notably, we observed that the intruder undergoes a size-ratio dependent streamwise velocity lag with respect to the bulk flow. Based on these observations, we propose a simple kinematic model which predicts the segregation duration of an intruder in bedload transport, incorporating a parameter representing the linear trajectory and a drift coefficient characterizing the streamwise velocity lag. Both parameters introduce a size ratio dependency based on our statistical analysis.

## I. INTRODUCTION

Morphological changes in rivers have a significant impact on landscapes, the quality of water resources, river ecosystems and public safety. Among the different processes involved,

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bedload plays a primary role in shaping the river as it accounts for the transport of the coarser particles at the river bed. Bedload formulas perform poorly in predicting sediment fluxes [1] and as a consequence bed morphology: one reason is grain size polydispersity, which leads to segregation. Particularly in mountain streams on steep slopes, the fluid shear stress can be sufficiently high to mobilize a wide range of particle size over several layers of grains within the bed. In this case, the grains can segregate vertically : the finer infiltrate the bed while the coarser move toward the top of the bed. This phenomenon was found to enhance sediment mobility [2, 3] and to participate in the creation of local bedforms and bedload sheet production [4, 5]. For now, bedload transport models at the river reach scale mainly consider hydrodynamic effects at the interface between the mean water flow and the granular bed. However, they ignore the granular scale phenomena acting inside the bed, which is needed to describe the vertical size segregation.

To account for this issue, Frey and Church [6, 7] urged the community to address bedload as a granular flow using the recent advances in the physics of granular media. Indeed, size segregation has been extensively studied in the context of dry granular flows [8–14]. The problem is classically seen as a competition between two mechanisms : a kinetic sieving mechanism, representing the infiltration of smaller grains into the interstices that form between the moving larger particles, and a "squeeze expulsion" mechanism whereby the finer grains push the larger particles upwards [8]. Several works have addressed kinetic sieving, based on the idea that small particles infiltrate because they are likely to be unsupported by the granular matrix [8, 15–17]. On the other hand, the mechanism responsible for squeeze expulsion is still a matter of debate [11, 18, 19].

Squeeze expulsion has been studied at the granular scale in various dry configurations [11, 14, 18–22]. These studies usually focus on the kinematic behavior or on the macroscopic vertical force usually defined as a lift force acting on a single intruder sphere, disk or cylinder plunged into a mono-dispersed granular bed. This force has been shown to depend on the vertical pressure gradient, vertical shear rate gradient and the size ratio between the intruder and the bed particles [11, 14]. Recently the model has been extended to mixtures of bi-dispersed material by Duan *et al.* [23] accounting for interactions between several intruders. Knowing expressions for the macroscopic forces acting at the grain scale, it is now possible to derive advection-diffusion models [24], a step towards the modeling of the squeeze expulsion at a larger scale.

Nevertheless, parametrizations for the segregation force are still missing. The effect of the size ratio on the segregation force has been investigated by Jing *et al.* [14] who proposed an empirical model for a single intruder in three different dry granular flows, confined either with a top wall or with a free surface. The model predicts a maximum segregation force when the intruder is two times the size of the bed particles, in agreement with Guillard *et al.* [11] for an intruder in a similar wall driven granular flow and with Trewthell *et al.* [22] for an intruder in an oscillatory shear cell. This result follows the maximum of segregation rate at a size ratio of 2 found by Golick and Daniels [10] in the case of a dry bi-dispersed mixtures of grains in a rotating annular cell. However, the physical origin of such an optimum is not yet clearly understood. The optimum value of two is not obvious either because the data in the different studies are usually quite scattered between 1.5 and 2.5.

In a bedload configuration, the squeeze expulsion has not been much studied. One reason lies in the exponential particle velocity and shear rate depth profiles [25] leading to two different granular flow regimes : the dense inertial regime in the flowing layer close to the bed surface, where particles undergo regular frictional contacts, and the quasistatic regime

deeper in the bed, where movements are less regular, making analysis more challenging due to the nonlocality of the flow rheology. In a bidispersed mixture configuration, Ferdowsi *et al.* [26] performed experiments on large particles segregating under laminar flows in an annular shear cell. The dynamics of such intruders was investigated for a size ratio equal to two and revealed that segregation occurred not only in the flowing layer but also in the quasi-static regime. However, the authors highlighted the necessity to perform experiments on segregation in less idealized configurations, particularly with a turbulent free surface flow. In this context, Rousseau *et al.* [27] presented the first experimental results for the trajectory of a spherical intruder initially immersed in a bedload flow. The authors focused on the phenomenological behavior of the intruder for different size ratios for two fluid flow configurations compatible with floods in mountain streams. Similar to Ferdowsi *et al.* [26], a vertical segregation has been observed in both the dense and the quasi-static flow regimes, with the intruder ascent being more random in the quasi-static regime compared to the dense regime. Interestingly, they found that the spatial trajectory of the intruder is much more intermittent below a size ratio close to 1.7 than above where the trajectory becomes linear with a slope independent of the size ratio. Additionally, they identified a potential minimum of segregation duration close to this size ratio value of 1.7. However, given the significant variability in their data, with only five to eight repetitions per configuration, robust conclusions were difficult to achieve. Furthermore, identifying the minimum segregation duration did not account for the intruder initial depth variability across repetitions. They suggested that a larger amount of repetitions should be conducted to investigate the problem more thoroughly.

This paper addresses these issues by providing a refined quantitative description of the intruder kinematics near the alleged optimum size ratio value of 2. For this purpose, we conducted 284 repetitions with six different size ratio configurations ranging from 1.25 to 3.0, and we adapted our methodology to account for the variability of the intruder initial position. A detailed description of the experimental set-up, protocol and explored configurations is given in Sec. II. This new extensive dataset represents a significant improvement over previous studies, enabling a statistical investigation of the intruder kinematics. In Sec. III, the spatial trajectory of the intruder is analyzed. Subsequently, in Sec. IV, the temporal vertical trajectory of the intruder is examined. Based on our observations, a simple kinematic model predicting the segregation duration of the intruder is introduced, incorporating size ratio dependent parameters  $\tan(\alpha)(r)$  and  $\beta(r)$  which quantify the linear trajectory of the intruder and the streamwise velocity lag of the intruder with respect to the bulk flow, respectively. Finally, conclusions are presented and perspectives on the modeling of the intruder dynamics and the effect of the size ratio are discussed.

## II. SETUP AND METHODS

### A. Setup

The experimental setup (Fig. 1) and equipment are the same as in Rousseau *et al.* [27]. It consists of a 2-meter-long and 15-millimeter-wide glass flume. The slope of the channel is  $S = 10 \pm 0.01\%$ , which corresponds to the case of a steep mountain stream. Water (density  $\rho^f = 1000 \text{ kg/m}^3$ ) flows from a constant head tank connected to the inlet of the flume. At the outlet, water is collected in a second tank and pumped back into the constant head reservoir. The constant water flow  $Q_w$  is manually controlled by a valve and measured

by an electromagnetic flow meter. The granular bed is made of spherical transparent glass beads (Sigmund Lindner type M) of density  $\rho^s = 2500 \text{ kg/m}^3$  and diameter  $d_s = 2 \pm 0.2 \text{ mm}$ , retained with a 60 mm weir at the outlet of the flume. Dried beads are injected at a constant rate  $Q_s$  with a K-Tron's Bulk Solids Pump mass controlled bead feeder in the upstream part of the channel and collected in a sieve at the outlet. The intruders are spherical black or red opaque polished glass beads (Sigmund Lindner type P) with the same density as the small beads.

A Photon Focus camera provided by Alliance Vision was positioned at the side of the flume to capture gray-level images of the region of interest (ROI) which was backlit with a LED panel delivering a uniform and constant brightness across all the images. The images were recorded at  $1024 \times 500$  pixels in grayscale at 260 frames per second. This frame rate corresponds to the displacement of one diameter of a small bead moving at fluid velocity above the bed surface between two images (Sec. II C). The scale of the images is  $0.212 \text{ mm/pixel}$ .

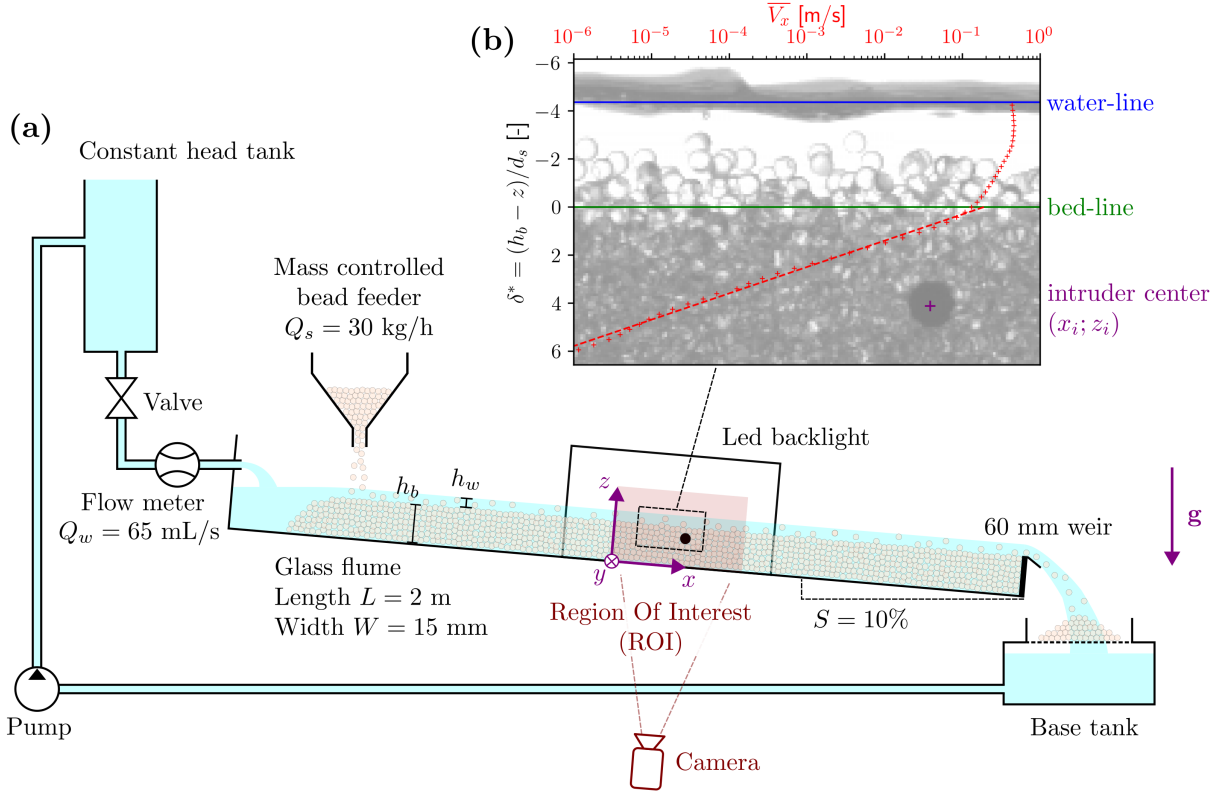


FIG. 1: (a) Experimental set-up (adapted from Rousseau *et al.* [27]). (b) Zoom on a camera snapshot with the elements detected with image processing (Sec. II C).  $\bar{V}_x$  is the velocity of the small beads averaged along the x-axis of the ROI.  $\delta^*$  represents the dimensionless depth below the bed surface where one unit represents the diameter of a small bead  $d_s$ .

## B. Protocol

The experiments were carried out at transport equilibrium meaning that on average the outlet sediment rate was constant and equal to the inlet sediment rate so that we had no evolution of the slope of the bed surface. In addition, the water flow rate was set so that at equilibrium, the slope of the bed surface was equal to the slope of the flume. To increase the number of repetitions, we performed experiments with up to three intruders distributed along the  $x$  axis of the ROI, sufficiently separated so as not to influence their dynamics. For each experiment, we performed the following steps:

(1) *Intruder placement.* The intruder center was placed manually at a target depth of four small bead diameters  $d_s$  below the bed surface. It was placed near the wall to be able to detect it with image processing. Due to the granular nature of the bed, placing the intruder at the same initial position between all the repetitions was one of the greatest difficulties we encountered. During this phase, we took care to minimize the disturbance we could introduce into the bed in the vicinity of the intruder. The method we used to place the intruder in the bed was the following. We started by digging a small hole with a plastic rod taking care not to compact the bed. This step was performed when the bed was wet but not flooded with water, which made the beads move easily and stick a little together so that they did not collapse in the hole while we dug. At this stage, the depth of the intruder was assessed using the mean bed surface vertical position from the previous experiment and measuring the vertical position of the intruder indirectly using a ruler on a camera snapshot of the ROI. Then the position of the intruder was adjusted until we reached a value close to the target  $4d_s$ . After that the hole was filled again with the beads previously left aside when we dug the hole. For some repetitions, it was easy to place the intruder at the target depth, but for others, the placement was more laborious and might have caused some disturbances within the bead arrangement surrounding the intruder. To overcome this problem and have the most comparable bed structures between the different repetitions, we delicately packed the bed with a large Plexiglas rod after placing the intruder, without pressing the beads down.

(2) *Experiment.* Once the intruders were placed into the bed, we opened the water valve to flood the bed but with no grain movement. Then we started the image acquisition and next, we turned on the bead feeder and the water valve to the target values. The acquisition was stopped when all the intruders were out of the ROI or when we run out of small beads. The bead feeder and water valve were then switched off. The bead feeder was always switched on and off just before the water valve to account for the bead travel time between the bead feeder and the flume. This allowed the bead and water fluxes to reach the bed almost simultaneously, reducing accretion or erosion phenomena when we started or stopped the experiments. This shortened the transient time to the equilibrium, increasing the chances for the intruder not to move too much before the bed and water surface stabilized.

(3) *Final check.* Just after each experiment, a 1 frame per second sequence of image was extracted and processed. This allowed us to check and discard any repetition for which the intruder had moved significantly during the transient phase to equilibrium, or had gone too far (more than  $1 d_s$ ) from the front wall during the experiment so that we could not detect it anymore. We considered the experiment to be in a steady state when the vertical evolution of the mean bed and water surfaces became second order compared to the initial depth of the intruder. For most of the experiments, the transient phase did not last longer than 60s as the bed was already set at equilibrium before the experiment. For the analysis,

we defined an initial time  $t_{init} = 65$  s, where we have added 5 s to account for the triggering of the water valve after the image acquisition started.

### C. Image processing

(1) *Intruder detection.* The intruder's opacity facilitated its detection among the transparent smaller beads. We used the detection part of the Matlab Bead Tracking library developed by Lafaye de Micheaux *et al.* [28]. The algorithm consists in thresholding the image to keep only black pixels, then computing the coordinates of the center of the pixel groups which have an area close to the mean area of the bead. This worked well when the intruder stayed closed to the wall during its ascent. Otherwise, the smaller beads overlapped in front of the intruder, making it blurred and more difficult to detect. This issue led to no detection when the intruder was getting closer to the bed surface for some of the retained repetitions over certain time steps. We were still able to retrieve the dynamics of the intruder in this zone by processing the 2600 images we got from the 10 last seconds of the acquisition. For common analysis of the intruder trajectory, we chose to process a lighter sequence of 5 frame per second over the whole experiment to reduce the processing time. After the detection phase, the coordinates of the center of the intruders  $(x_i; z_i)$  were assigned to the right intruder dividing the image into a grid of  $n$  cells, corresponding to the number  $n$  of intruders placed into the ROI. Then a Savitsky-Golay filter was used on all the coordinates to remove some noise due to the detection algorithm.

(2) *Bed and water surface detection.* To obtain the position of the bed surface and the water free surface (bed and water lines in our 2D images), we used the algorithms developed by Lafaye de Micheaux [29] and Frey *et al.* [30]. For the bed line, the process uses a Sobel gradient to detect the changes in pixel intensity and a marker-based watershed transformation to bring out the shape of the line. The processing parameters were previously tuned to position the bedline at the interface between the bed with beads in permanent contacts and the layer of saltating beads above. For the water line, the approach is to employ a morphological closing with a structuring element of a height close to the one of the water line, then extracts its shape with another marker-based watershed transformation. The mean vertical position of the bed and water lines were obtained by averaging in space along the entire width of the ROI and in time from  $t_{init}$  to the end of the acquisition, on the same 5 frames per second sequence as for the intruder detection. This allowed us to get rid of the fluctuations of the surfaces. The water height  $h_w$  was then computed as the difference between the waterline and the bedline. The bed elevation  $h_b$  is equal to the position of the mean bed line since the bottom of the ROI matches the bottom of the flume.

(3) *Velocity field of the bed.* The velocity of the small beads was computed using the Lucas-Kanade optical flow like technique used by Rousseau and Ancey [31, 32]. It appeared to be more suited for our case than both classical particle image velocimetry techniques which often fail for strong velocity gradient, and particle tracking techniques which perform poorly with overlapping dense particles. From the computed velocity field, we were able to retrieve and fit the exponential vertical velocity profile down to  $5d_s$  below the bed surface, using two sequences of 15000 images covering the almost same  $\approx 60$  s period: one with a small time step of  $1/260$  s to get information on the upper part of the bed (for  $\delta^* < 4$  and  $V_x > 1e^{-4}$  m/s) and another with a large time step of  $1/10$  s to extract the velocities deeper

in the bed. The fitted velocity profile averaged over all the repetitions is

$$\overline{V}_x(\delta^*) = V_0 e^{-\frac{\delta^*}{C}}, \quad (1)$$

where  $\delta^* = (h_b - z)/d_s$  is the dimensionless depth of the intruder inside the bed of height  $h_b$ ,  $V_0 = 0.189$  m/s is the mean streamwise velocity of the small grains at the top of the bed and  $C = 0.488$  is a constant. The velocity profile is plotted in Fig. 1(b).

#### D. Experimental parameters and configurations

The bed was built with beads of diameter  $d_s = 2 \pm 0.2$  mm and the sediment transport rate was set to  $Q_s = 30$  kg/hr to reproduce the high shear stress configuration of Rousseau *et al.* [27]. The corresponding water flow rate to have equilibrium with the bed surface parallel to the bottom of the flume was experimentally found to be  $Q_w = 65.0$  ml/s, the same as Rousseau *et al.*. The corresponding water depth  $h_w = 8.7$  mm was computed over all the experiments (Sec. II C). The mean velocity of the fluid was then  $U = Q_w/(Wh_w) = 0.50$  m/s. The flow was turbulent (bulk Reynolds number  $Re = 4UR_h/\nu \approx 8000$  where  $R_h = 4.0$  mm is the hydraulic radius and  $\nu = 10^{-6}$  m<sup>2</sup>/s is the kinematic viscosity of the water) and supercritical ( $Fr = U/\sqrt{gh_w} = 1.7$  where  $g = 9.81$  m/s<sup>2</sup> is the acceleration of gravity). The Shields number is  $Sh = \rho^f S R_{hb}/((\rho^s - \rho^f)d_s) = 0.25$  where  $R_{hb} = 7.4$  mm is a corrected hydraulic radius which takes into account the fluid dissipation distribution between the bed and the walls [33]. The Rouse number  $Ro = w_s/(\kappa\sqrt{gR_{hb}S}) = 8.2$  (where  $w_s$  is the settling velocity computed with the Ferguson and Church [34] formula and  $\kappa = 0.41$  is the von Kármán constant) ensures there is only bedload and no suspension.

The size of the intruder  $d_l$  was the only controlling parameter to change between the experiments. Values were chosen to analyze the effects of the size ratio  $r = d_l/d_s$  close to  $r = 2$ . We ended up with the following configurations and number of repetitions.

| Intruder diameter $d_l$ [mm] | Size ratio $r = d_l/d_s$ | Number of repetition |
|------------------------------|--------------------------|----------------------|
| 2.5                          | 1.25                     | 49                   |
| 3.0                          | 1.5                      | 45                   |
| 3.5                          | 1.75                     | 50                   |
| 4.0                          | 2.0                      | 50                   |
| 5.0                          | 2.5                      | 45                   |
| 6.0                          | 3.0                      | 45                   |

TABLE I: Summary of the different configurations varying the diameter of the intruder.

As shown in Tab. I, we chose to favor the number of repetitions over the number of configurations to strengthen our results given the dispersion of the intruder dynamics showed by Rousseau *et al.* [27]. All our experiments were conducted at the same Shields number  $Sh = 0.25$  for which we can afford placing the intruder deep enough into the bed while keeping a relatively low experiment duration and remaining representative of the shear stress encountered in a natural mountain stream. Indeed, the deeper we place the intruder, the less the relative standard deviation of its initial depth between all the repetitions.

Although we targeted the same initial depth of  $4d_s$  for all the intruders, the resulting depth range we managed to get was of the order of the diameter of the small beads, i.e.

from  $3.5d_s$  to  $4.5d_s$  for 81% of the repetitions (Fig. 2). The initial intruder depths are quite symmetrically distributed around the median, which is equal to the target  $4d_s \pm 0.1d_s$  for all the configurations, apart for  $r = 2$  for which the intruder was often placed closer to the bed surface owing to inherent difficulties in the placement process. The dispersion is mainly related to the discrete nature of the granular bed, which acts like a grid requiring the intruder to fit within. It can also be imputed to our inability to precisely anticipate the position of the bed surface during the intruder placement.

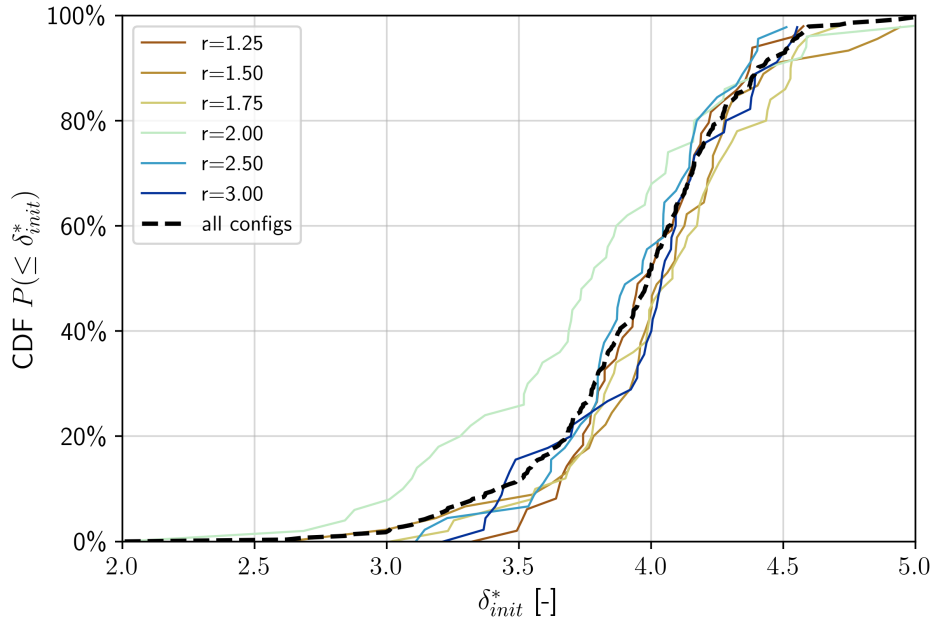


FIG. 2: Initial intruder depth distribution for each configuration (colored lines) and for all configurations combined (dashed black line).  $\delta^* = (h_b - z)/d_s$  is the dimensionless depth of the intruder inside the bed of height  $h_b$ . The  $y$  axis shows the percentage of repetitions for which the initial depth of the intruder is equal or below the  $x$  axis referred depth.

### III. SPATIAL TRAJECTORY OF THE INTRUDER

This first result section deals with the spatial trajectory of the intruders. First we give a qualitative analysis focusing on the raw detected coordinates of the intruders. Then we discuss more quantitatively the results on the slope of the fitted linear spatial trajectory.

#### A. Qualitative analysis

Fig. 3 shows the spatial trajectory of the intruders for each configuration. For all the repetitions, the intruder globally rises from its initial position to the bed surface ( $\delta^* = 0$  in Fig. 3). Locally, it either rises or sinks intermittently during its ascent, but it never moves upstream. The ascent is more and more irregular as the size ratio decreases. Indeed, when the diameter of the intruder is of the same order as the surrounding beads, the intruder is more likely to fit into the interstices of the granular matrix. On the contrary, when the size ratio increases, the intruder is less sensitive to the discrete nature of the granular bed.

Also, as observed in [27], the deeper the intruder the more irregular its ascent. However, the initial depth of the intruder (color scale in Fig. 3) does not seem to affect the global shape of the trajectory. For  $r \geq 2$ , the trajectory of the intruder appears to be quite linear with the same slope whatever the size ratio. For  $r < 2$ , as  $r$  decreases, the trajectory is less linear and the intruder travels more and more in the streamwise direction.

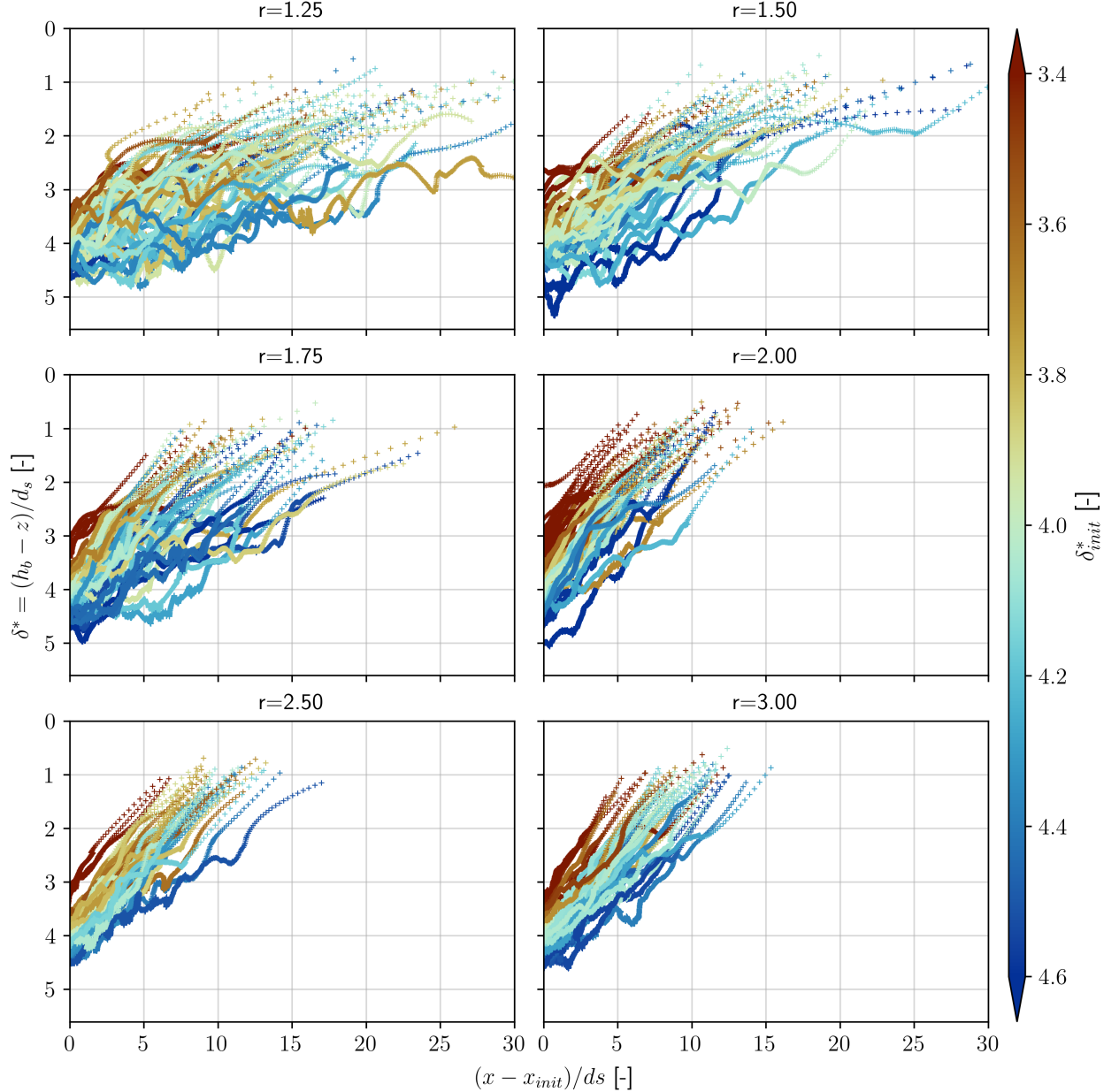


FIG. 3: Spatial trajectory of the intruder for each repetition for the different size ratios. The  $x$  axis represents the streamwise displacement of the intruder shifted from its initial  $x$  coordinate. The  $y$  axis shows the depth evolution of the intruder inside the bed of height  $h_b$ . Both axes are scaled by the small particle diameter  $d_s = 2$  mm. The initially deeper intruders are shown in cold colors whereas the initially higher intruders are drawn in warm colors according to the colorbar.

To have a better view on the dispersion of the trajectories between the different repetition of each configuration, we plotted in Fig. 4 the relative spatial displacement of the intruders from their initial position. A linear trajectory was fitted on each repetition. The data points for which  $\delta^* < 1.5$  were not taken into account in the regression. Above this depth, the top of the intruders might protrude above the local bed surface and the trajectory might be modified by collisions with saltating beads. The data points are not equally distributed. The deeper the intruder, the more concentrated the data points, due to the global acceleration of the intruder. Therefore, to be the most representative of the linear spatial trajectory, the best way to do the regression was found to be a weighted least square regression with respect to the kernel density estimation of the  $x$  coordinate, forcing the linear fit to cross the initial position  $(x_{init}; z_{init})$  of the intruder.

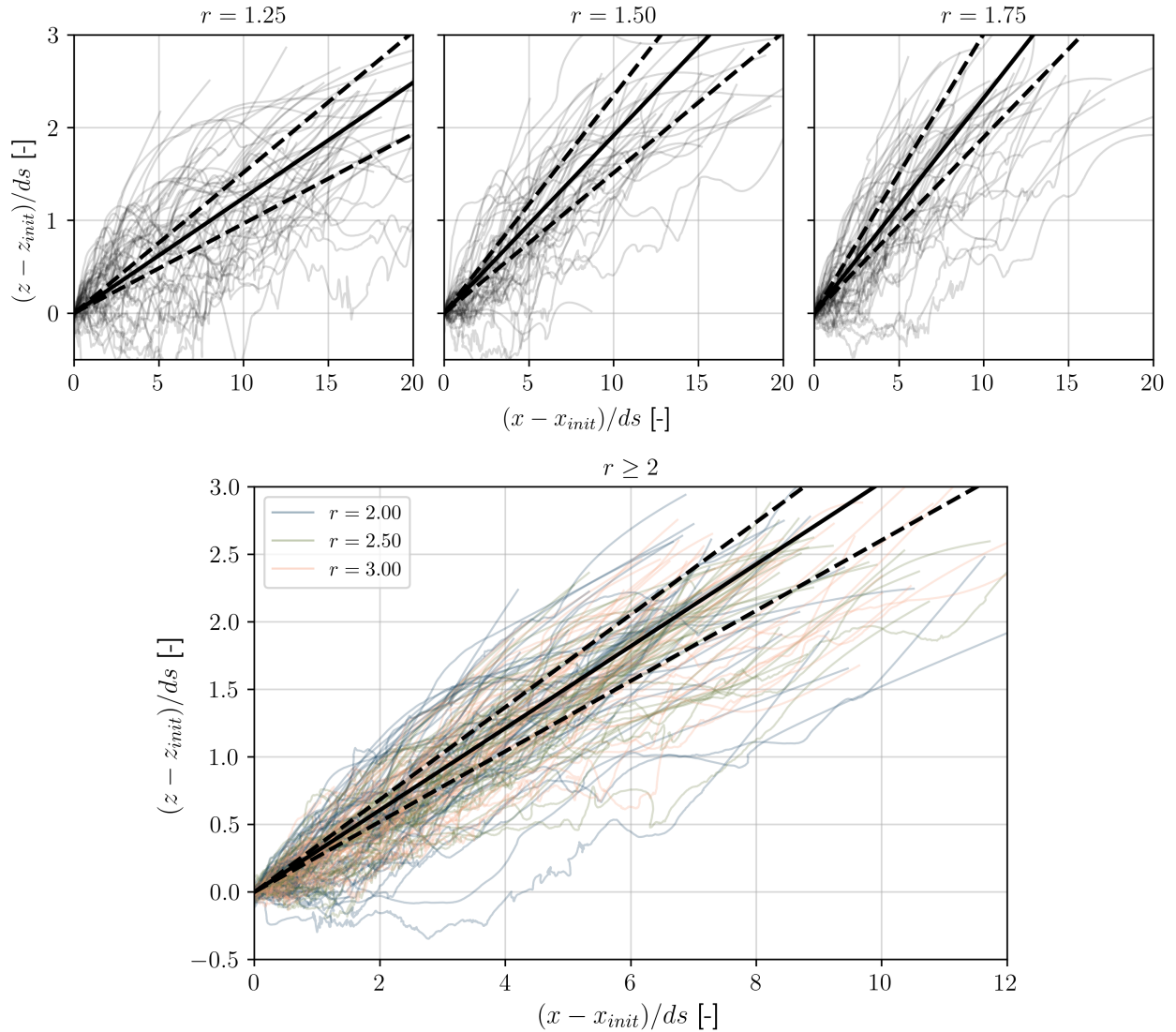


FIG. 4: Spatial displacement of the intruders from their initial position. The black lines show the median and the quartiles of the distribution of the fitted linear spatial trajectory.

As shown in Fig. 4, the evolution of the slope of the fitted linear spatial trajectories for  $r < 2$  increases with  $r$ , while for  $r \geq 2$  all the trajectories tend to collapse as shown by the well-mixed colors corresponding to each size ratio.

### B. Distribution of the slope of the linear spatial trajectories

The distribution of the slope of the fitted linear spatial trajectory is presented in Fig. 5 along with the data from Rousseau *et al.* [27].

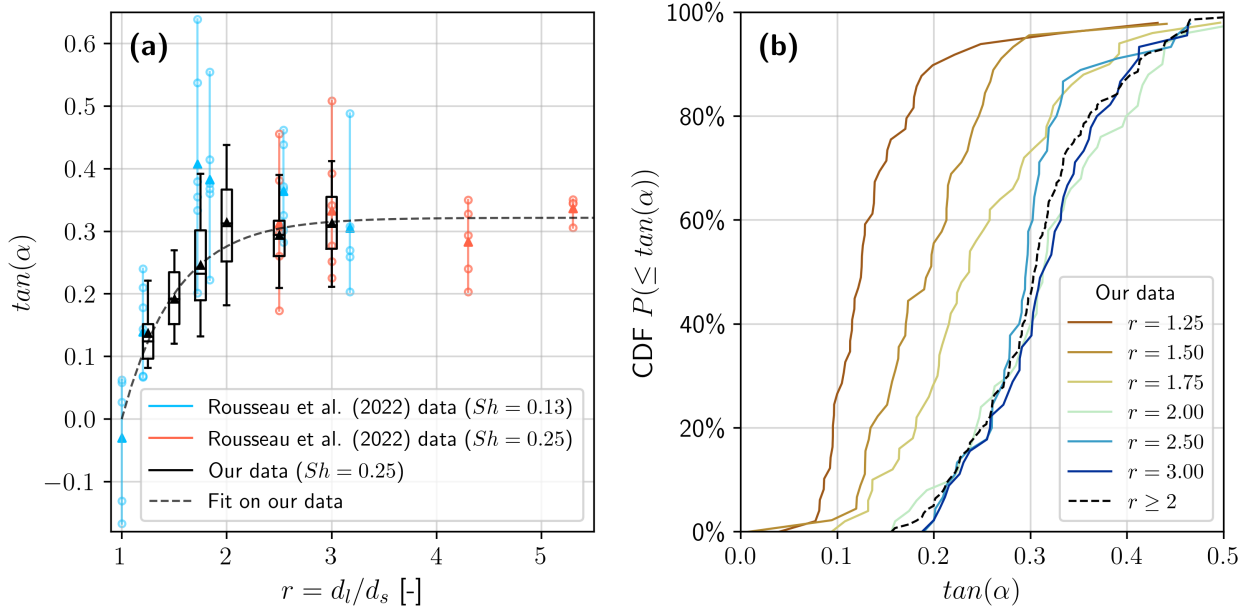


FIG. 5: **(a)** Aggregation of our data and the one from Rousseau *et al.* [27], showing the slope of the fitted linear spatial trajectory  $\tan(\alpha)$  function of the size ratio  $r = d_l/d_s$ . The boxplots include all the values inside the 5th and 95th percentiles, the boxes represent the inter-quartiles and the horizontal bars are the medians. The circles are the values from Rousseau *et al.* [27]. The triangles are the mean values. The dashed black curve represents the equation  $\tan(\alpha)(r) = 0.32 \left(1 - e^{1.95(1-r)}\right)$  with coefficients 0.32 and 1.95 mean square fitted to the medians of our data ( $r^2 = 0.94$ ). **(b)** Cumulative density function of the slope of the fitted linear spatial trajectory  $\tan(\alpha)$  for each size ratio in our experiments.

From a statistical point of view, we clearly see in Fig. 5(a) that the slope of the trajectory decreases between  $r = 2$  and  $r = 1.25$  and we could expect it to go toward zero for  $r = 1$  corresponding to a passive tracer behavior. For  $r \geq 2$ , the slope reaches a plateau value close to 0.3 ( $\alpha = 16.7^\circ$ ). The data of Rousseau *et al.* [27] at  $Sh = 0.25$  [red in Fig. 5(a)] are consistent with our data at  $r = 2.5$  and  $r = 3$ , which allows us to extend our conclusions at  $r = 4.3$  and  $r = 5.3$ , confirming that there is no evolution of the slope for  $r \geq 2$ . Moreover we find the same increasing slope behavior for  $r < 2$  as Rousseau *et al.* found in a lower shear stress configuration with  $Sh = 0.13$  [blue in Fig. 5(a)].

The high number of repetitions of our data allows us to analyze more thoroughly the distribution of the slope of the fitted linear trajectory [Fig. 5(b)]. The distributions are quite symmetric around the median for  $r \geq 2$ . Looking at the combined distribution of the

size ratios 2, 2.5 and 3 [black dashed line in Fig. 5(b)], we note that the median slope is 0.31. We saw in Fig. 3 that as  $r$  decreases, the intruder is more likely to navigate through smaller holes opened within the small beads. Strangely we would have expected a larger dispersion as  $r$  decreases, but we see in Fig. 5(b) that this is not the case as the slope of the CDF around the median is even decreasing between  $r = 1.25$  and  $r = 1.75$ . However we do observe for  $r < 2$  a longer tail as  $r$  decreases making the distributions not symmetric around the median. Also, we point out that the dispersion is larger for  $r = 1.75$  than for the other size ratios.

#### IV. TIME VERTICAL TRAJECTORY OF THE INTRUDER

The results presented in this section deal with the time vertical trajectory of the intruders. In the same manner as in the previous section, we first give a qualitative description from the detected coordinates of the intruder. Then we discuss in more detail the effect of the size ratio, including it in a simple kinematic model.

##### A. Qualitative analysis

Fig. 6 shows the time vertical trajectory of the intruders for each configuration. The intruder globally accelerates as it moves upward, which is consistent with the non linear velocity profile of the small beads (Eq. (1)). Looking closer at the shape of the time vertical trajectories on Fig. 6, two zones may be distinguished: an irregular trajectory where the intruder is advected horizontally by the granular flow, and a smoother ascent towards the bed surface which looks very similar between all the repetitions. Moreover, on the one hand looking at the color scale on Fig. 6 we clearly see that the initial depth has a significant impact on the time taken by the intruder to reach the bed surface. Whatever the size ratio, deeper intruders (cold colors in Fig. 6) take generally more time to reach the bed surface than intruders initially placed higher in the bed (warm colors). On the other hand, we do not observe any significant differences in terms of segregation duration between the different size ratios.

As all the trajectories seem to have the same shape in the final ascent of the intruder, it suggests taking the final time as a reference in the following part of the analysis. By doing so, we avoid the influence of the initial depth differences between all the repetitions, giving us better insights on how the depth evolves with time and what is the effect of the size ratio in this evolution. The resulting shifted collapsed trajectories are shown on Fig. 7. The semilog scale in Fig. 7 shows clearly that the depth evolves exponentially with time. This evolution is demonstrated further in Sec. IV C. The effect of the size ratio does not appear clearly due to the relatively high dispersion of the curves and requires a more detailed analysis of the distribution of the trajectories (Sec. IV B).

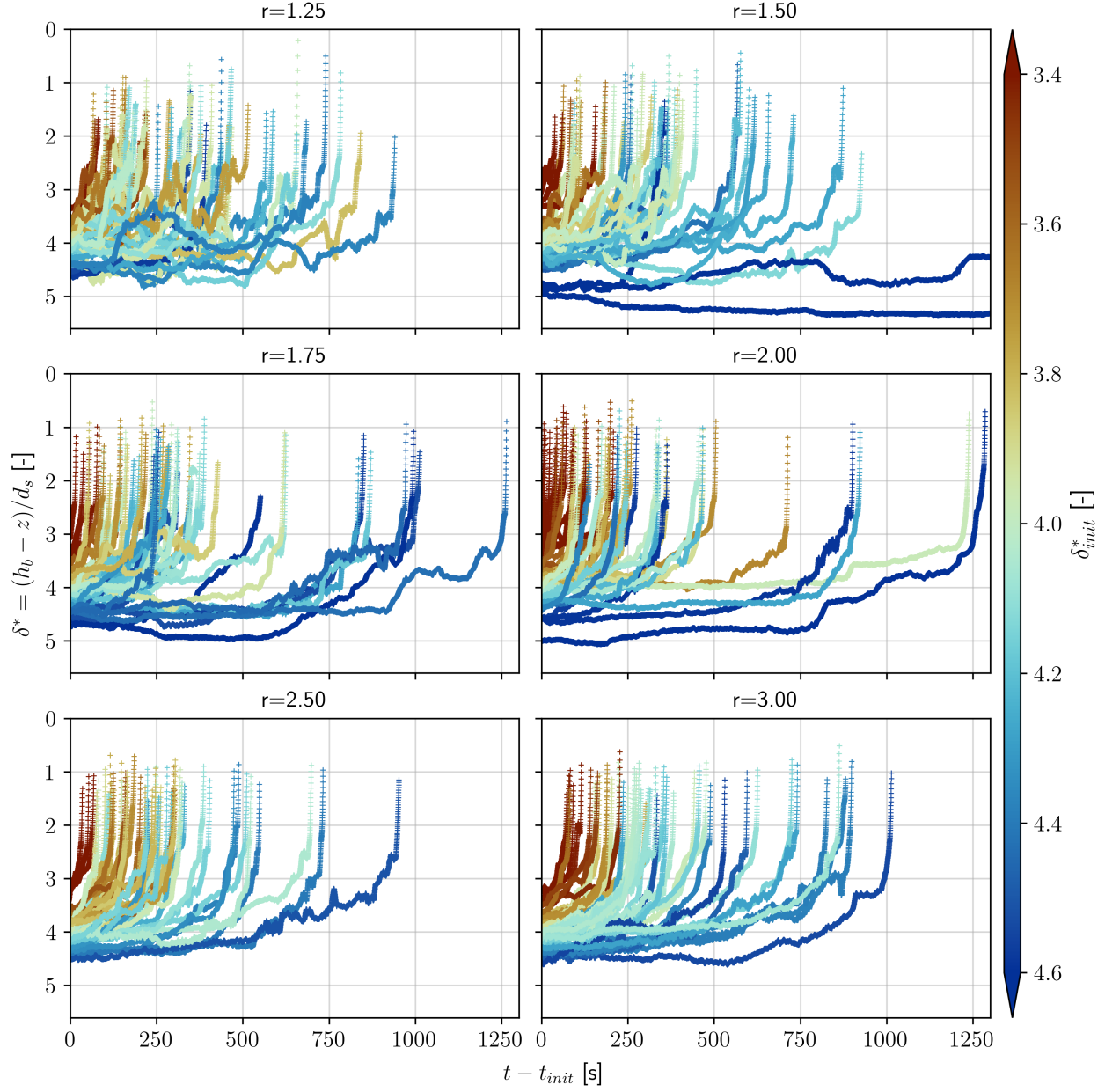


FIG. 6: Time vertical trajectory of the intruder for each repetition for different size ratios. The  $x$  axis represents time after the transient phase of duration  $t_{init}$ . The  $y$  axis shows the depth evolution of the intruder scaled by the small particle diameter  $d_s = 2$  mm (same as Fig. 3). The initially deeper intruders are shown in cold colors whereas the initially higher intruders are drawn in warm colors according to the colorbar. For  $r = 1.50$  the two deeper trajectories in blue are cut since they reached the bed surface (0 on the  $y$  axis) in 1754s and 3074s respectively.

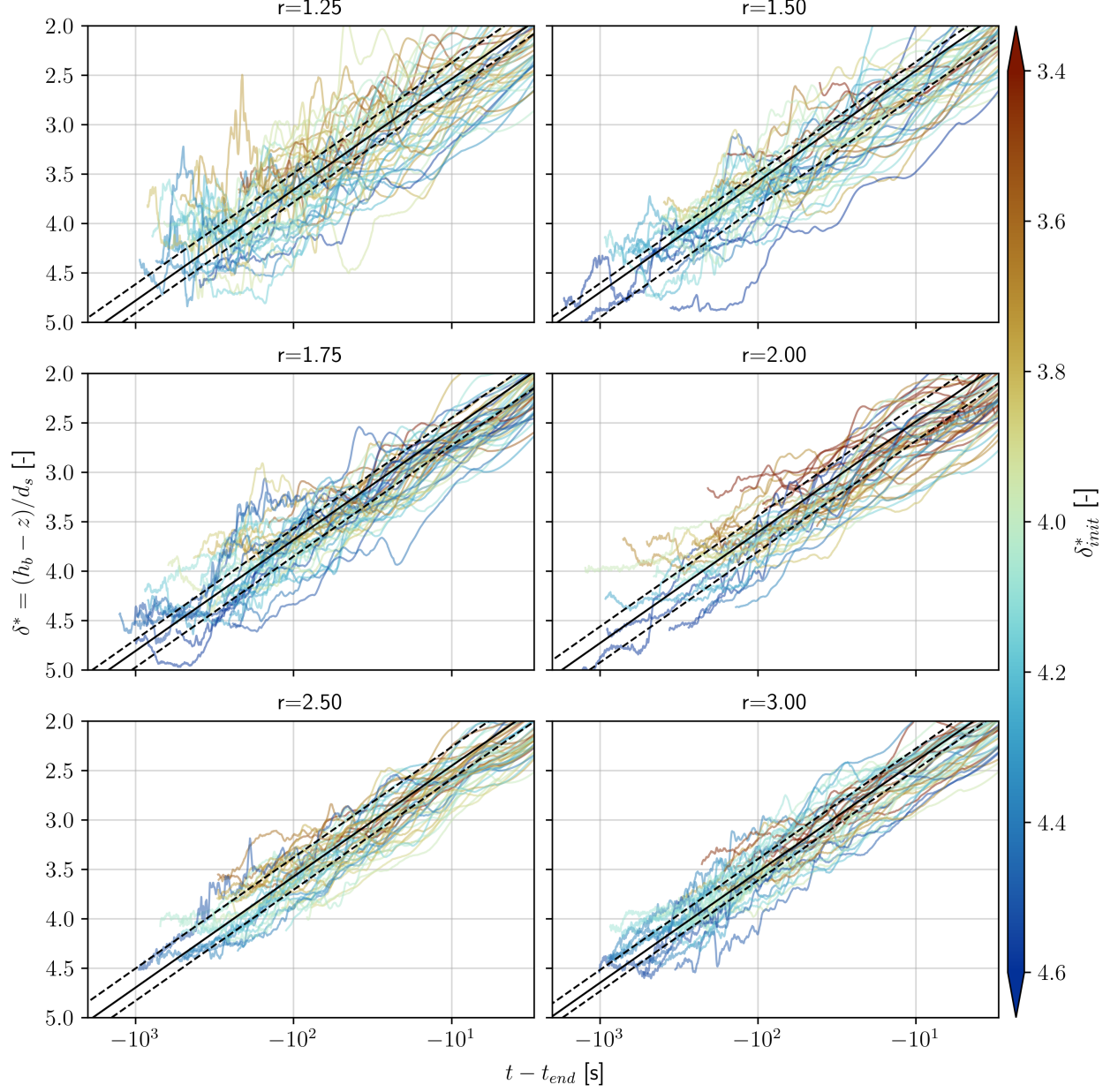


FIG. 7: Time vertical trajectory of the intruder where the  $x$  axis has been shifted from  $t_{end}$ , the time when the intruder reaches the bed surface. The  $x$  axis can be interpreted as the remaining ascent time to reach the bed surface. The  $y$  axis shows the depth evolution of the intruder inside the bed of height  $h_b$  scaled by the small particle diameter  $d_s = 2$  mm (same as Figs. 3 and 6). The initially deeper intruders are shown in cold colors whereas the initially higher intruders are drawn in warm colors according to the colorbar. The black lines (median and quartiles) represent the equation  $t - t_{end} = A \left( e^{\delta_{init}^*/C} - 1 \right)$  where  $A$  has been fitted for each repetition and  $C$  is the constant from Eq. (1) (see Sec. IV C).

### B. Effect of the size ratio on the segregation duration

To extract information about the size ratio effect on the time vertical trajectory, we chose to look specifically at the distribution of the remaining ascent time  $\Delta t = t_{end} - t(\delta^*)$  from different depths (Fig. 8).  $t(\delta^*)$  is taken as the time when the intruder crosses the corresponding depth  $\delta^*$ , allowing us to include in the remaining ascent time  $\Delta t$  the periods when the intruder position stagnates near the target  $\delta^*$ . Our data allow us to conduct the analysis in the depth range  $\delta^* \in [3.2; 4.1]$ . Indeed, for  $\delta^* < 3.2$ , the relative uncertainty on  $\Delta t$  becomes too high due to really small  $\Delta t$  values, smaller than 10 seconds. For  $\delta^* > 4.1$ , the number of repetitions decreases drastically, from approximately 40% of the repetitions remaining at  $\delta^* = 4.1$  to less than 10% at  $\delta^* = 4.5$  which is not sufficient to discuss the statistics. Fig. 8 shows that in the range  $\delta^* \in [3.2; 4.1]$ , whatever the depth, the remaining ascent time increases with the size ratio  $r$  for  $r \geq 2$ . For  $r < 2$ , the dispersion of the data globally increases as  $r$  decreases. Regarding this dispersion in the range  $\delta^* \in [3.2; 3.8]$ , we cannot say much about how  $\Delta t$  evolves with  $r$  for  $r \leq 2$ , although if we focus on the mean behavior (black triangles), we see that  $\Delta t$  globally decreases with  $r$  making  $r = 2$  a minimum. In the case  $\delta^* = 4.1$ , we observe a clearer trend with  $\Delta t$  decreasing with  $r$  for  $r \leq 2$ .

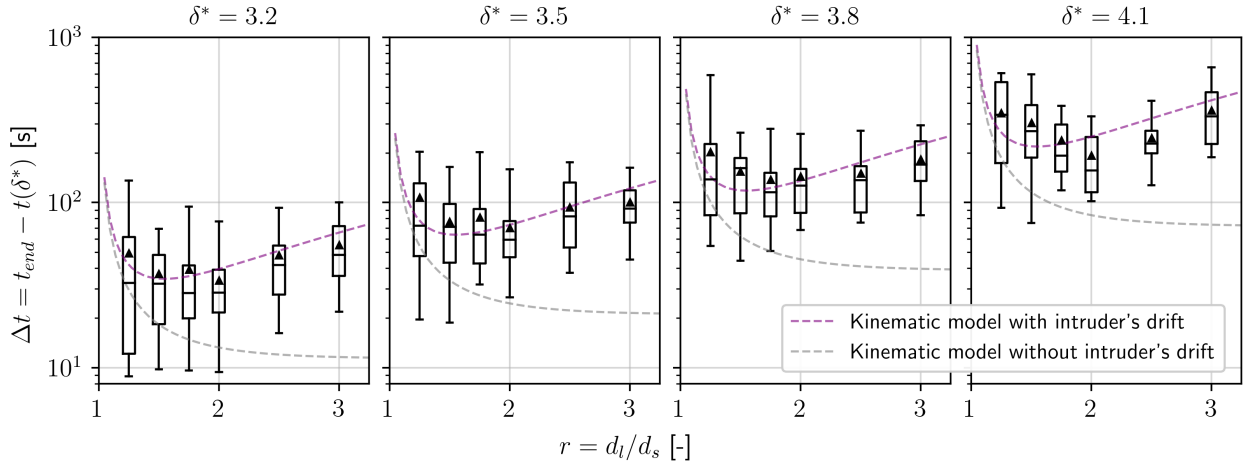


FIG. 8: Distribution of the intruder remaining ascent time or segregation duration  $\Delta t$  starting from four different depths and for each size ratio. The boxplots include all the values inside the 5th and 95th percentiles, the boxes represent the quartiles, the horizontal bars are the medians and the triangles the means. The dashed lines represent the kinematic models developed in Sec. IV C: the purple curve (Eq. (7)) accounts for a streamwise velocity drift observed between the intruder and the surrounding particles, while the gray curve (Eq. (5)) assumes no drift.

The data of Rousseau *et al.* [27] at  $r = 2.5$  and  $r = 3$  were plotted in Fig. 9. For this comparison to be effective, the depths of the intruders in the Rousseau *et al.* data were reevaluated considering the average bed surface on the entire experiment instead of the one at the beginning of the experiment to be consistent with the protocol adopted herein. The remaining ascent time was compared from  $\delta^* = 3.2$  for which we still get at least three repetitions in Rousseau *et al.* while remaining deep enough to neglect relative uncertainties on the ascent time. Both datasets are consistent which allows us to extend our conclusions to higher size ratio: the segregation duration increases with the size ratio for  $r \geq 2$ .

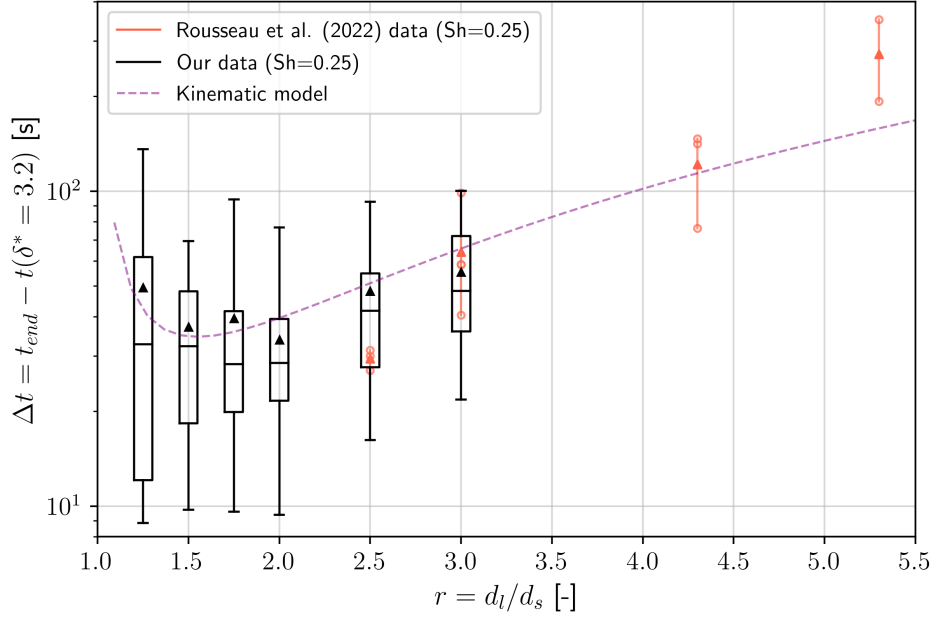


FIG. 9: Aggregation of our data and those of Rousseau *et al.* [27], showing the remaining ascent time  $\Delta t = t_{end} - t$  of the intruder from  $\delta^* = 3.2$  as a function of size ratio  $r = d_l/d_s$ . The boxplots include all the values inside the 5th and 95th percentiles, the boxes represent the interquartiles and the horizontal bars are the medians. The red circles are the values from Rousseau *et al.*. The triangles are the mean values. The dashed line represents the kinematic model developed in Sec. IV C, Eq. (7).

These results are compatible with Golick and Daniels [10], Guillard *et al.* [11] and Jing *et al.* [14, 21], who found respectively an increasing segregation velocity for the first one and an increasing segregation force for the others for  $r > 2$ . However, we do not find an extremum at  $r \approx 2$  like they did because regarding the distribution of our data, it is not clear if  $\Delta t$  increases with  $r$  for  $r < 2$ . Nevertheless, considering our results in term of the intruder spatial trajectory, we do find a transition in the behavior of the intruder at  $r = 2$ .

The large number of repetitions in our data set allows us to look at the cumulative density function of the remaining ascent time at different depths in Fig. 10 where our data present a quite symmetric distribution in semilog scale. The order of magnitude of the ascent duration is the same for all size ratios for each depth. However, the size ratio seems to have an effect on the dispersion of the data. The lower the size ratio (warmer colors), the higher the dispersion. This is consistent with the time trajectory looking like staircase shapes for low size ratios, with sometimes the intruder getting trapped in the interstices between the small beads.

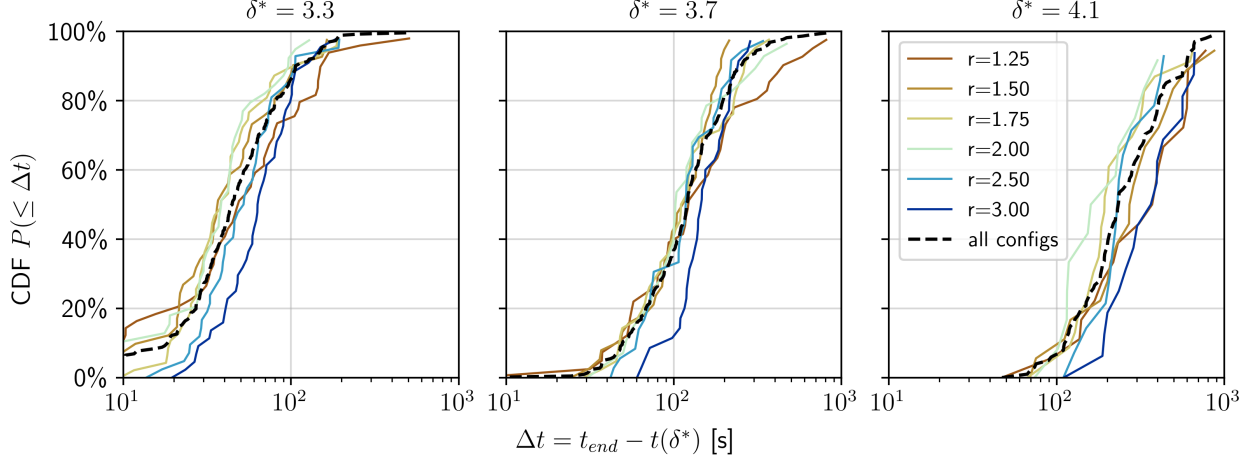


FIG. 10: Cumulative density functions of the remaining ascent time of the intruders from three different depths and for each size ratio.

### C. Kinematic model of the time vertical trajectory

To model the ascent of the intruder, a simple kinematic model is introduced. Based on our observations, the spatial trajectory of the intruder  $(x_i, z_i)$  can be considered as linear at first order with a slope  $\tan(\alpha)$  depending on the size ratio  $r$ . Hence we can write

$$dz_i = \tan(\alpha(r))dx_i. \quad (2)$$

Deriving against time, we get

$$\frac{dz_i}{dt} = \tan(\alpha(r))\frac{dx_i}{dt}, \quad (3)$$

where  $\frac{dx_i}{dt}$  is the streamwise velocity of the intruder.

Let us assume the streamwise velocity of the intruder is equal to the streamwise velocity of the bed. Then we would get

$$\frac{dx_i}{dt} = \bar{V}_x = V_0 e^{-\frac{\delta_i^*}{C}}, \quad (4)$$

where  $V_0$  and  $C$  are the fitted parameters obtained from Eq. (1) and  $\delta_i^* = (h_b - z_i)/d_s$ . Then, the ascent time is obtained as follows:

$$\Delta t = \int_t^{t_{end}} dt = \int_{z_i}^{h_b} \frac{-1}{\tan(\alpha(r))V_0} e^{\frac{h_b - z_i}{Cd_s}} dz_i = \frac{Cd_s}{\tan(\alpha(r))V_0} \left( e^{\frac{\delta_i^*}{C}} - 1 \right). \quad (5)$$

In Eq. (5), the  $C$  and  $V_0$  parameters were obtained from the optical flow measurements presented in Sec. II C. The dependency of  $\alpha$  on the size ratio  $r$  was obtained from the tracking of the intruder and presented in Fig. 5.

We plotted the prediction of Eq. (5) as a dashed gray line in Fig. 8. It does not match our data at all, especially as the size ratio increases. For us, the explanation is that the larger the intruder, the greater it drifts on the smaller beads. It can be observed in the video in

the Supplemental Material [35]: the intruder is much slower than the surrounding beads in the streamwise direction. To model this behavior, we assume that the streamwise velocity of the intruder is proportional to the mean velocity of the bed particles at the same depth  $\overline{V}_x$  with a coefficient  $\beta$  called the drift coefficient which may depend on the size ratio  $r$ .

$$\frac{dx_i}{dt} = \beta(r) \overline{V}_x. \quad (6)$$

Assuming that the  $\beta$  coefficient is constant over depth, we can obtain the ascent time the same way as in Eq. (5):

$$\Delta t = \frac{C d_s}{\tan(\alpha(r)) \beta(r) V_0} \left( e^{\frac{\delta^*}{C}} - 1 \right). \quad (7)$$

To compute  $\beta$  for each repetition, Eq. (7) has been fitted to the data points from Fig. 7, knowing  $C$  and  $V_0$  from Sec. II C and  $\tan(\alpha)$  from Sec. III B. We note that in Fig. 7, the slope  $1/C$  of the fitted curves in semilog scale well matched our data. This emphasizes the importance of the bed velocity profile in modeling the intruder time vertical trajectory, since the constant  $C$  characterizes the exponential bed velocity profile.

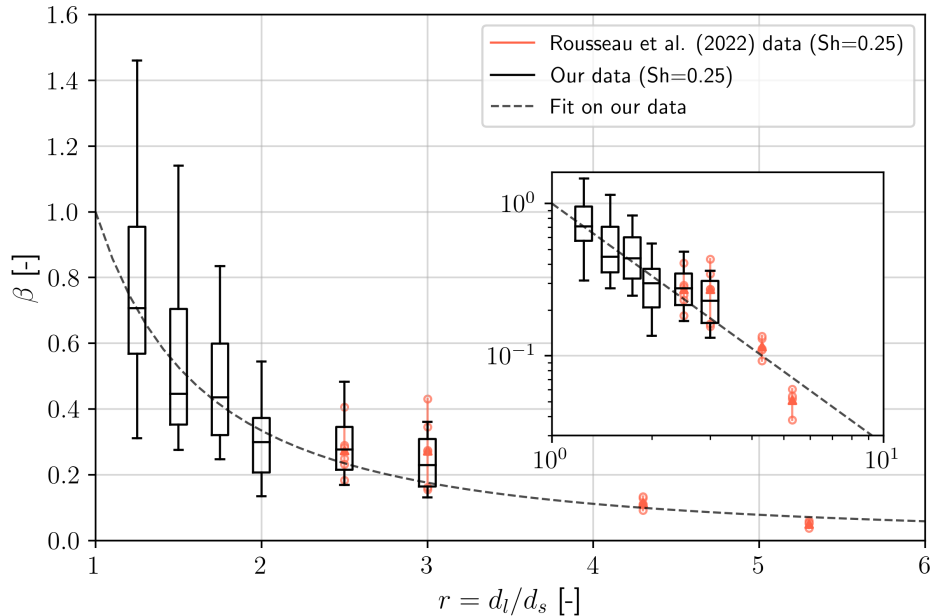


FIG. 11: Distribution of the drift coefficient  $\beta$  (the ratio of the intruder streamwise velocity to the mean streamwise velocity of the bed at the same depth) against the size ratio  $r$ . The boxplots include all the values inside the 5th and 95th percentiles, the boxes represent the quartiles and the horizontal bars are the medians. The red circles are the values from Rousseau *et al.* [27]. The triangles are the mean values. The dashed black line represents the equation  $\beta(r) = r^{-1.58}$  with coefficient 1.58 mean square fitted on the medians of our data ( $r^2 = 0.92$ ). The inset figure displays the same data on a log-log scale.

The computed  $\beta$  values are shown in Fig. 11 for both our dataset and the one of Rousseau *et al.* [27] in the high Shields configuration. We also conducted a fitting analysis on the medians of our data, opting for a simple power law due to its nearly linear behavior on a log-log scale. Notably, the data from Rousseau *et al.* [27] align well with our own and with

our fitted power law. The  $\beta$  evolution is consistent with  $\beta$  to tend to 1 as  $r$  reaches 1 since the intruder would then behave like a passive tracer. Also, the decreasing evolution of  $\beta$  with  $r$  is compatible with the case of an intruder in a liquid flow for which the greater the size of the intruder, the greater the drag forces and the slower the intruder. From this simple analogy, we could use our  $\beta$  coefficient to link our kinematic model to a dynamic model, for which the expression of the forces are needed. This is beyond the scope of the present work and is left as a perspective.

It is worth mentioning that the velocity lag between the intruder and the bulk bed flow has previously been reported by Van Der Vaart *et al.* [19] using discrete element simulations in a different configuration of dry avalanche on an  $22^\circ$  (40,4%) inclined plane with a rough fixed bottom. In their paper they provide an expression for the velocity lag  $V_{lag} = V_{xi} - V_x = a(1/r - 1)/\eta$ , with  $a$  a parameter fitted on their data and  $\eta$  the granular viscosity. Based on their expression, the drift coefficient would be expressed as  $\beta = 1 + a(1/r - 1)/(\eta V_x)$  which is consistent with our observations since  $\beta = 1$  when  $r = 1$  and  $\beta$  decreases with  $r$ .

For a practical use of the kinematic model, we recall the empirical expression of  $\tan(\alpha)(r)$  and  $\beta(r)$  which were fitted on the medians of our data (see Figs. 5 and 11):

$$\tan(\alpha)(r) = 0.32 \left( 1 - e^{1.95(1-r)} \right), \quad (8)$$

$$\beta(r) = r^{-1.58}. \quad (9)$$

We can assess the prediction of the model with the purple dashed line in Eq. (7). The model with the drift fairly predicts the size ratio dependency, contrary to the model without drift. Looking closer, the predicted values seem a bit overestimated and the experimental transition value at the size ratio of 2 is not obtained, which we attribute to the approximations introduced by the  $\tan(\alpha)$  and  $\beta$  fits. The present kinematic model still allows us to approximate the remaining ascent time of an intruder of a specific size ratio from a specific depth. It is mainly valid for size ratios larger than 2 for which the fluctuations of the trajectories are small. For  $r < 2$  the model is less deterministic and would require including a statistical representation of the fluctuations.

## V. CONCLUSION AND PERSPECTIVES

We conducted laboratory experiments on a large particle segregating upwards in bedload transport. The dependency of the size ratio on the intruder dynamics has been investigated for six different size ratios between 1.25 and 3. For each size ratio, around 50 repetitions have been performed resulting in approximately 300 individual trajectories. This allowed us to draw robust conclusions regarding the spatial and temporal trajectory of the intruder. First, we observed a regime transition for size ratios between 1.25 and 2: as the size ratio increases, the data show less variability and the slope of the spatial trajectory gets steeper. Then, above this regime transition, for size ratios larger than 2, the spatial trajectory of the intruder is linear with a slope close to 0.3 and the time for the intruder to reach the bed surface increases.

Using these observations, we derived a simple kinematic model to express the segregation duration as a function of the size ratio. To fit the data, a drift coefficient has been introduced, representing the streamwise velocity lag between the intruder and the surrounding bed particles. Based on our experimental results the velocity lag depends on the size ratio.

The experimental dataset presented in this work could serve as a guideline to develop and validate a discrete element approach of the same configuration. This numerical model would give access to contact forces, particle concentration, particle rotation and a more detailed bed velocity field. Using this complementary approach, we could better quantify the physical mechanisms at work during the observed regime transition. In addition, the drift coefficient could be linked to physical forces such as the drag or lift forces calculated directly from the discrete contact forces.

Another interesting point we could address with the discrete element model is the influence of the sidewalls on the intruder dynamics. It should also be pointed out that index matching techniques could be used as well to address the sidewall issue.

We believe that all these introduced elements would allow proposing a precise mechanical model for the behavior of a single large particle in bedload transport.

The data set is available in Zenodo [36]. A GIF showing the ascent of the intruder is available in the Supplemental Material [35].

## ACKNOWLEDGMENTS

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